

Proto Board: Op-Amp Gain-Staged Input with BJT Clamp

Single-channel prototype of the Rev 2 input circuit for real-world testing.

Uses the Rev 1 PCB for power supply and THAT 1246 differential receiver.

Parts List (all THT)

Ref	Value	Part	Notes
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U1	Op-amp	TL072CP (DIP-8)	Use one half. OPA1656 is SMD only – TL072 is fine for clamp testing (JFET input, $\pm 15V$, DIP-8).
R_in	10k	1/4W metal film	
R_fb	25k pot	16mm panel pot or trim pot	Linear taper for testing
R_out	2.2k	1/4W metal film	
Q1	2N3906	T0-92 (EBC pinout)	PNP positive clamp
Q2	2N3904	T0-92 (EBC pinout)	NPN negative clamp
R11, R14	10k	1/4W metal film, 1%	

Reference divider high side

	R12, R13		1k		1/4W metal film, 1%				
Reference divider low side									

	C5, C6		470uF/25V		Electrolytic, radial				
Reference filtering									

	R_led		1k		1/4W				
LED current limit									

	LED		Red 3mm		Standard				
Clip indicator									

	C3, C4		100nF		MLCC ceramic			0p	
-amp decoupling, one per supply pin									

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Connections to Rev 1 Board

Power (3 wires):

- +12V analog supply
- -12V analog supply
- AGND

Signal in:

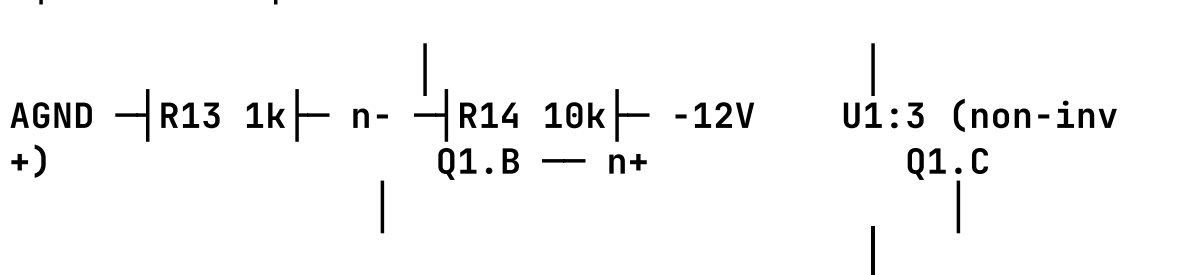
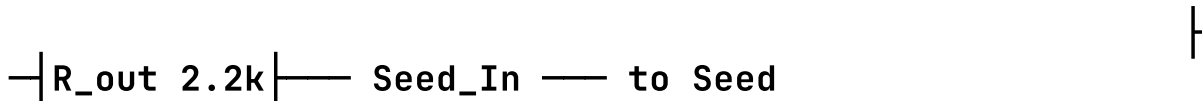
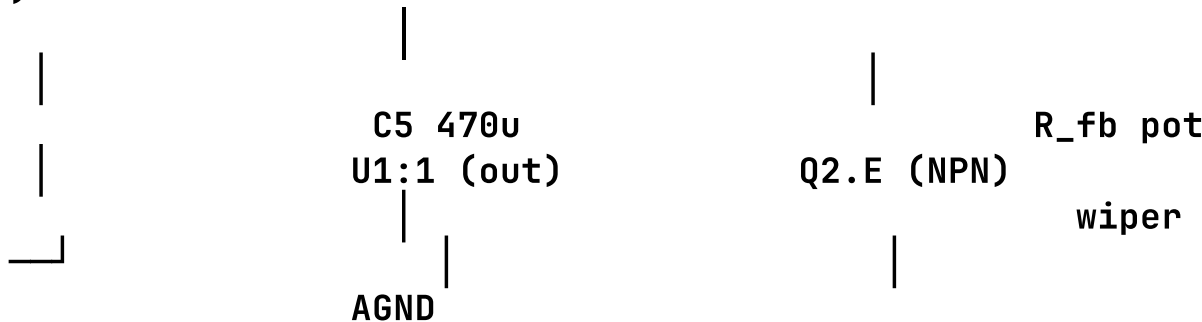
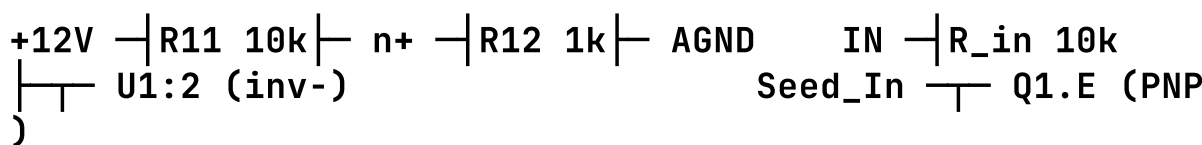
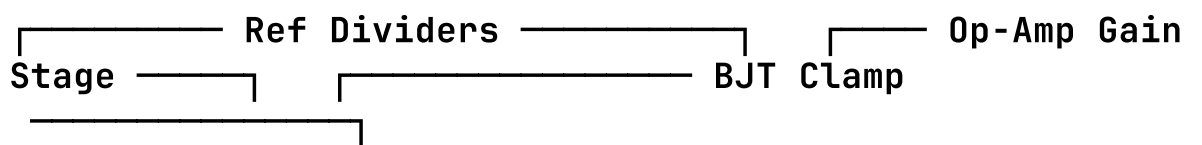
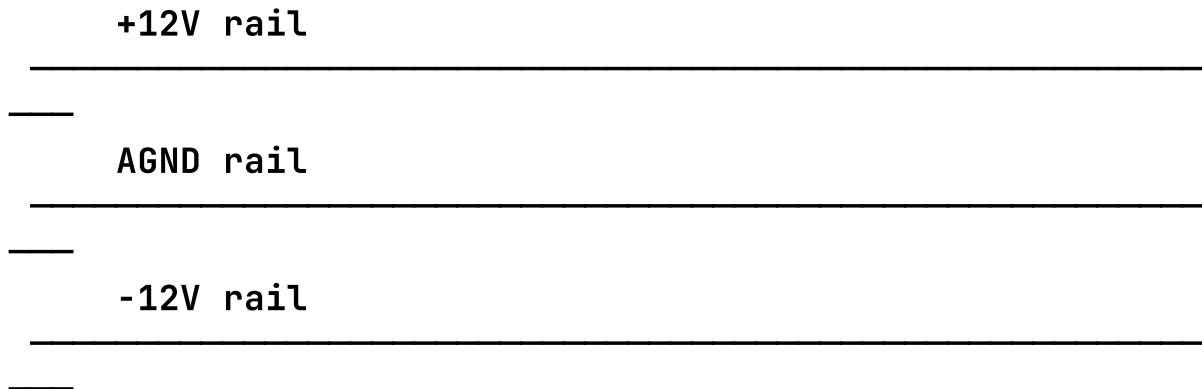
- THAT 1246 output -> proto board "IN"

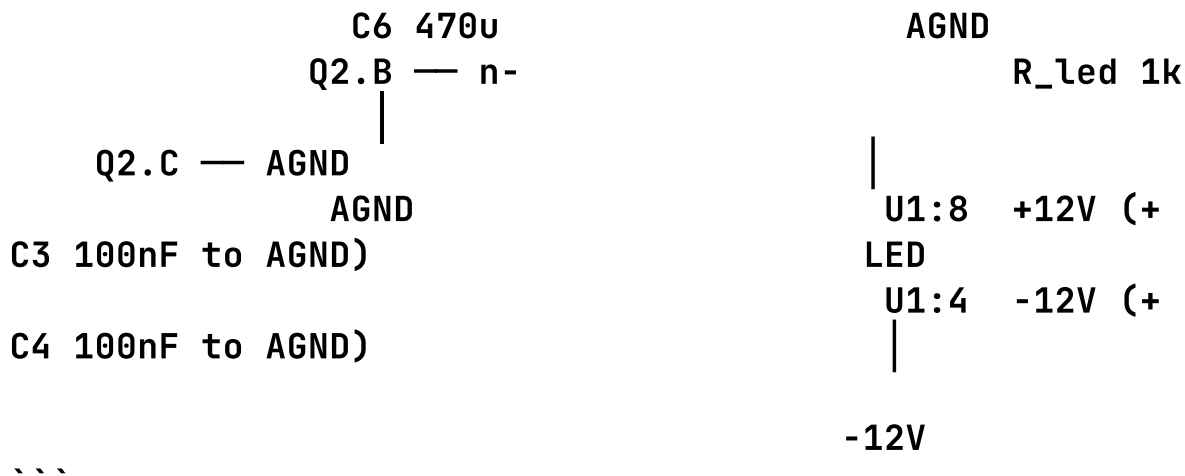
Signal out:

- Proto board "Seed_In" -> Daisy Seed audio input pin (pin 21 or 22 for L/R)

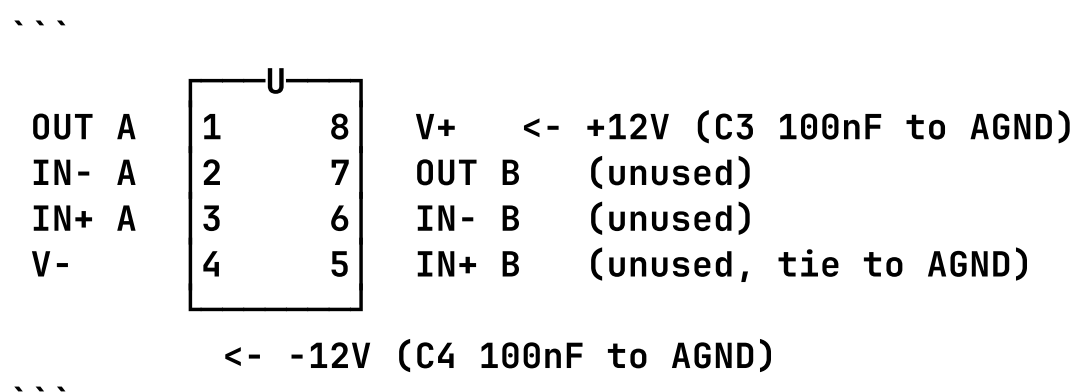
Breadboard Layout (single channel)

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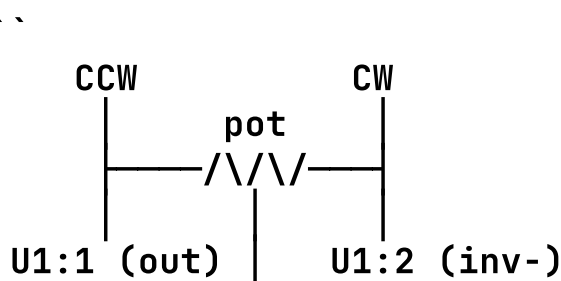


Op-Amp Wiring (TL072CP, DIP-8)



- Pin 1 (OUT A): to R_fb pot CCW terminal + R_out
- Pin 2 (IN- A): to R_in + R_fb pot wiper
- Pin 3 (IN+ A): to AGND
- Pin 5 (IN+ B): tie to AGND (prevents unused half from oscillating)

R_fb Pot Wiring



wiper
(also to U1:2)

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- CW terminal: to U1 pin 2 (inverting input) and R_in
- CCW terminal: to U1 pin 1 (output)
- Wiper: to U1 pin 2 (inverting input)

This way, turning the pot CW increases R_fb (more gain). At full CCW the wiper shorts to the output, giving minimum gain.

BJT Wiring (T0-92, flat side facing you, pins down)

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2N3906 (Q1, PNP):			2N3904 (Q2, NPN):		
E	B	C	E	B	C
Seed	n+	R_led->LED->-12V	Seed	n-	AGND
_In		(clip indicator)	_In		

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Both emitters connect to the same Seed_In node. This is the critical junction where R_out, both BJT emitters, and the wire to the Seed pin all meet.

Key Nodes

Node	DC voltage	What connects here
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+12V	+12V	R11, U1 pin 8, C3
-12V	-12V	R14, U1 pin 4, C4, LED
cathode (via R_led)		

AGND	0V	R12, R13, C5, C6, U1 pin
3, U1 pin 5, Q2.C, C3, C4		
n+	+1.091V	R11/R12 junction, C5+, Q1
.B		
n-	-1.091V	R13/R14 junction, C6-, Q2
.B		
IN	signal	THAT 1246 output, R_in
inv-	virtual ground	R_in, R_fb wiper, U1 pin
2		
U1 out	signal	R_fb CCW, R_out
Seed_In	signal (clamped)	R_out, Q1.E, Q2.E, wire
to Seed pin		

Test Points

- **n+, n-**: verify +/-1.091V DC with multimeter
- **U1 pin 1**: op-amp output – should swing freely below clamp, saturate at ~+/-11V under overdrive
- **Seed_In**: signal after clamp – scope here to observe clamp knee and hard limiting
- **LED**: visual clip indication – lights when positive clamp conducts